Daniela Jacqueline Zúñiga Jiménez

Agroecologist · Soil Science & Environmental Data Modelling

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EDUCATION

Bachelor's Degree in Agroecology 2021 – Present
Universidad Autónoma de Yucatán (UADY) — *Mérida, Yucatán, Mexico*

> **Thesis:** "Machine learning-based pedotransfer functions for estimating agriculturally relevant physicochemical properties in soils of Yucatán."

Specialization: PTF modelling · machine learning applied to soil science · spatial data analysis · tropical soil systems (karst environments).

RESEARCH EXPERIENCE

Research Internship — Modelling of Ammonia Emission from Field Crop Production 2026
AgroParisTech Grignon Experimental Farm & UMR ECOSYS — Grignon (78), France · *Supervisor: Dr. Sophie Carton*

- Selected and processed monitoring data (meteorological variables, soil characteristics and crop-management practices) from the Trajectoire agronomic platform.
- Applied dedicated modelling programs to estimate NH_3 emissions from nitrogen fertilization across seven cropping-system management strategies, analyzing and synthesizing the results.
- Collaborated with the experimental-farm R&D team and the UMR ECOSYS research group, specialized in ammonia-volatilization modelling.

Research Internship — Soil Properties Across Physiographic Provinces of Mexico 2025
Universidad Nacional Autónoma de México (UNAM) & LUGA · *Supervisor: Dr. Francisco Bautista*

- Built machine-learning pedotransfer functions to estimate bulk density, soil organic carbon and cation exchange capacity (CEC) from INEGI physicochemical soil databases.
- Trained and compared predictive models in Weka and Python, evaluating algorithm performance to select the most reliable estimation approach per property.
- Developed reproducible Python workflows for cleaning, processing and analyzing national-scale soil datasets across contrasting physiographic provinces.

Research Internship — Microbial Diversity in Agroecosystems of Yucatán 2025
Universidad Autónoma de Yucatán (UADY) · *Supervisor: Dr. Miriam Ferrer*

- Characterized soil fungal communities from karst sascab substrates and plant rhizospheres by ITS metabarcoding, processing 8 samples into 918 amplicon sequence variants (ASVs) with the QIIME 2 pipeline.
- Assigned taxonomy and quantified community composition (Ascomycota-dominated, led by Acremonium and Fusarium) and computed alpha diversity (Shannon, observed richness) and rarefaction curves in R (phyloseq).
- Compared shared and unique taxa across substrates and color morphotypes using Venn diagrams and abundance bar plots.

ACADEMIC & TECHNICAL EXPERIENCE

Research Project — Agroecological Systems in Transition (Kantunil, Yucatán)

- Conducted socio-economic assessments of rural production systems.
- Performed statistical analysis using Jamovi and Excel, and built dashboards and visualizations in Power BI.
- Carried out spatial analysis using QGIS.

DIPLOMAS & PROFESSIONAL TRAINING

Centro de Altos Estudios en Geomática
Python Applied to Spatial Data Science · Cloud-Based Spatial Analysis with Python & Google Colab · Spatial SQL Applied to Spatial Data Science

Pontificia Universidad Católica de Chile
Data Analytics for Agricultural Management

Asociación Mexicana de Estudios del Karst
Soil Science Applied to Land Evaluation in the Mexican Tropics

Ancora Academy
Environmental Impact Assessment Procedures and Risk Analysis

Global Standards
FSSC 22000 V6 · HACCP Alliance (certification-system implementation)

TECHNICAL SKILLS

Data Science & Programming: Python (spatial & environmental data), SQL (Spatial SQL), Google Colab, R, Weka.

Geospatial & Analytical Tools: QGIS, Power BI, Microsoft Excel (Advanced), Jamovi.

Methodologies: Machine learning for environmental modelling, pedotransfer-function development, soil physicochemical analysis, spatial data analysis, environmental impact assessment, statistical analysis and estimation.

LANGUAGES

Spanish — Native **English** — C1 **French** — B1 **Arabic** — A2